

VF179_V1_R3_P332BEV

Functional Description

332BEV

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1 Rear Parking Assistance [VF179_V1_R3]

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1.1.1 Vehicle Function Area

Driving Assistance

1.1.2 Vehicle Function Group

Parking Assistance

1.1.3 Vehicle Function Owner

Laura D'Alessandro

1.2 *Revision Notes*

Date	Author	Description
2018/09/21	Laura D'Alessandro	First release Carry-Over document from: N.A. Derived document from: VF179_V1_R8_P520MY

2019/02/22	Laura D'Alessandro	CR12965 UPDATE DESCRIPTION - Impacted Component: PAM 1) Paragraph: Functional Diagram Changes: - Changed the internal signal <i>PAM_CmdIgnSts.Info</i> to <i>PAM_OperationalModeSts.Info</i> (RAR 249) - Removed the signals <i>PARK_INFO.Graphic_XHR</i> - Removed the input signal <i>CI_IGW1.ShiftLeverPosition</i> and added the signal <i>TRANSM2.ShiftLeverPosition</i> - Added the internal signal <i>InternalLightSts</i> and added external input VF041 (RAR-461 alignment) - Added the input signal <i>BRAKE4.VehicleStandStillSts</i> 2) Paragraph: Variable Definition Changes: - Changed the KEY ON and KEY OFF variables (RAR 249) - Added Standstill Condition 3) Paragraph: PAM State Machine Changes: - Added KEY OFF State Description (RAR 249) - Changed KEY ON State Description (RAR 249, RAR 304) - PAM ON_Active Substate Description: added a filtering time in standstill condition 4) Paragraph: Visual Alert Table Changes: - Removed the signals <i>PARK_INFO.Graphic_XHR</i> in the table - Added Multiple Obstacles visualization Management 5) Paragraph: PAM Chime Requirements Changes: - Added Multiple Obstacles-Chime Management 6) Paragraph: Diagnosis and Recovery Changes: - Added diagnosis ID 19.0 (missing message) and ID 20.0 (plausibility) 7) Paragraph: Configuration Parameters Table Changes: - PAM_STUCK_TIMEOUT increased to 15 s - T_Filter_Alerts parameter added 8) RAR alignments (RAR 3772, 3773, 3775, 3776, 3777, 3778) - Impacted Component: BCM 1) Paragraph: Functional Diagram Changes: - Added the internal signal <i>BCM_OperationalModeSts.Info</i> 2) Paragraph: PAM button and LED Management Changes: - Changed the internal signal <i>BCM_CmdIgnSts.Info</i> to <i>BCM_OperationalModeSts.Info</i> 2) Paragraph: Gateway Management Changes: - Removed the gateway of the signals <i>PARK_INFO.Graphic_XHR</i> - Removed the gateway of the signal <i>TRANSM2.ShiftLeverPosition</i> - Removed the gateway of the signal STATUS_TELEMATIC.AUDIOSts_Telematic
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		• Impacted Component: IPC 1) Paragraph: Functional Diagram Changes: • Changed the internal signal <i>IPC_CmdIgnSts.Info</i> to <i>IPC_OperationalModeSts.Info</i> (RAR 3480) • Added the input signal <i>STATUS_TELEMATIC.AudioSts_Telematic</i> • Added the indication "PARKING ASSISTANCE-VEHICLE-GRAPHIC.V2" 2) Paragraph: IPC Algorithm Requirements Changes: • Added Ignition Working Conditions (RAR 3480) • Updated the "Display Activation and Deactivation" paragraph • Updated internal signals • Updated "IPC - Chime Activation and Deactivation" paragraph 3) Paragraph: Indication Management Changes: • Indication "PARKING ASSISTANCE-VEHICLE-GRAPHIC.V2" added in order to handle the graphic layout activation • changed conditions to trigger the USER ENABLED and USER DISABLED indications (RAR 3480) 4) Paragraph: Configuration Parameters Table Changes: • Added T_PAMSts_valid parameter • Added ARC_FLASH_DUTY_CYCLE and ARC_FLASH_PERIOD values • Impacted Component: PAM 1) Paragraph: Diagnosis Table and Recovery Description Changes: • Updated Working Conditions (RAR 248) • Changed the signal <i>PAM_OperationalModeSts.Info</i> in Diagnosis 5.0 (RAR 248) • Removed the diagnosis on the <i>BH_IGW1.AUDIOSts_Telematic</i> signal • Impacted Component: BCM 1) Paragraph: Diagnosis Table and Recovery Description Changes: • Updated Working Conditions • Impacted Component: IPC 1) Paragraph: Diagnosis Table and Recovery Description Changes: • Updated Working Conditions (RAR 3480) • Added diagnosis for the missing message of <i>STATUS_TELEMATIC</i>
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2019/05/17	Laura D'Alessandro	CR13239 UPDATE DESCRIPTION - Impacted Component: PAM 1) Paragraph: PAM State Machine Changes: - Conditions to store the internal variables MODE_PAM and PAMFault updated. These variables are now stored at every change of state [RAR 925] 2) Paragraph: Configuration Parameters Changes: - TReverseGear parameter increased to 400 ms [RAR 303] - Impacted Component: IPC 1) Paragraph: IPC Algorithm Requirements Changes: - IPC internal signals: removed the setting of the signal <i>STATUS_PAM.RearSensorSts</i> equal to "Not_Active_Failure", because this code is never used in the IPC logic [RAR 4295] 2) Paragraph: Indication Management Changes: - Added the PAM_Configuration PROXI parameter condition into the indication PARKING ASSISTANCE SENSOR BLOCKED.V1 [RAR 1161] - PARKING ASSISTANCE.V1: deactivation condition corrected 3) Paragraph: Diagnosis Table and Recovery Description Changes: - ID 4.0 Description: resuming condition after a fault added [RAR 1041] - ID 7.0 Description: heal condition updated 4) Paragraph: Configuration Parameters Changes: - T_PAMSts_valid parameter updated with the correct unit of measure [RAR 1042, 4294] - T_Telematic parameter added [RAR 4349] - Impacted Component: LSS 1) Paragraph: LSS Requirements Changes: - Switch Acquisition requirement corrected
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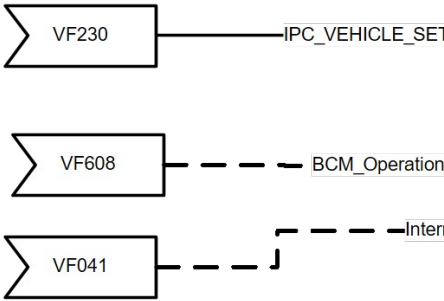
This document is based on template "VF Template 9.2"

1.3 Introduction

The system warns the driver of obstacles located behind the vehicle during the parking manoeuvre. It assists the driver during parking by sensing obstacles also situated outside the driver's field of view.

The information regarding the presence and distance of an obstacle is transmitted to the driver in the form of acoustic and visible signals. The acoustic signal may be pulsed at a rate depending on distance.

1.4 Functional Diagram



1.5 External Interfaces

1.5.1 External Inputs

VF400

VF608

VF230

VF651

VF200

VF041

1.5.2 External Outputs

VF651

1.6 Control Unit

1.6.1 ECU

IPC

Logic

Actuation

PAM

Logic

Sense

BCM

Gateway

1.6.2 EMCU

LSS

1.6.3 External Device

1.6.4 Additional Component

IPC

1.7 I/O

Park_Assist.Light

Park_Assist_On_Off.Switch

Trailer_Hook.Sensor

IPC_Display.GUI

R_Bumper_Left_Parking.Sensor

R_Bumper_Central_Parking.Sensor

R_Bumper_Right_Parking.Sensor

1.8 Signal

1.8.1 Hardwire

Trailer.Info

LedControlSts.Req

Park_Assist_On_Off.Req
R_Bumper_Central_Parking.Info
R_Bumper_Left_Parking.Info
R_Bumper_Right_Parking.Info

1.8.2 Internal

ChimeActivation_LHR
ChimeActivation_RHR
ChimeRepetitionRate_Rear
ChimeType_Rear
PAMSystemFault
RearSensorSts
Park_Assist_Light.Cmd
PAMSystemSts
Alert_CHR
Alert_LHR
Alert_RHR
DisplayActivation
PamAlertMode
PAMAboveSpeed
Park_Assist_On_Off_Switch.Req
IPC_OperationalModeSts.Info
PAM_OperationalModeSts.Info
BCM_OperationalModeSts.Info
InternalLightSts

1.8.3 B/BH-CAN

PARK_INFO.Alert_CHR

This signal indicates which arc shall be displayed in the central hand, rear position of the car graphics.

PARK_INFO.Alert_LHR

This signal indicates which arc shall be displayed in the left hand, rear position of the car graphics.

PARK_INFO.Alert_RHR

This signal indicates which arc shall be displayed in the right hand, rear position of the car graphics.

PARK_INFO.ChimeActivation_LHR

This signal indicates the chime activation request for the left hand, rear audio speaker, or rear hardwired buzzer.

PARK_INFO.ChimeActivation_RHR

This signal indicates the chime activation request for the right hand, rear audio speaker, or rear hardwired buzzer.

PARK_INFO.ChimeRepetitionRate_Rear

This signal indicates the repetition rate for the rear audio speakers.

PARK_INFO.ChimeType_Rear

This signal indicates the Chime Type for the rear audio speakers.

PARK_INFO.DisplayActivation

This signal indicates the request to display the car graphics layout on the IPC/EVIC screen.

STATUS_PAM.PAMSystemFault

This signal indicates if the PAM is in a faulted condition

STATUS_PAM.RearSensorSts

This signal indicates the activation state of the rear sensors

STATUS_PAM.PAMSystemSts

This signal indicates the activation state of the PAM

IPC_VEHICLE_SETUP.PamAlertMode

This signal is used to give information about the PAM Alert Mode selected

STATUS_TELEMATIC.AUDIOSts_Telematic

Used to give information about Telematic Module audio availability

STATUS_PAM.PAMAboveSpeed

This signal is used to manage a indication, related to the deactivation of PAM system, when the vehicle speed is too high.

1.8.4 LIN

1.8.5 C-CAN1

1.8.6 C-CAN2

BRAKE1.VehicleSpeedVSOSig

It is the average vehicle speed calculated.

BRAKE1.VehicleSpeedVSOSigFailSts

It is the average vehicle speed fail status.

PARK_INFO.Alert_CHR

This signal indicates which arc shall be displayed in the central hand, rear position of the car graphics.

PARK_INFO.Alert_LHR

This signal indicates which arc shall be displayed in the left hand, rear position of the car graphics.

PARK_INFO.Alert_RHR

This signal indicates which arc shall be displayed in the right hand, rear position of the car graphics.

PARK_INFO.ChimeActivation_LHR

This signal indicates the chime activation request for the left hand, rear audio speaker, or rear hardwired buzzer.

PARK_INFO.ChimeActivation_RHR

This signal indicates the chime activation request for the right hand, rear audio speaker, or rear hardwired buzzer.

PARK_INFO.ChimeRepetitionRate_Rear

This signal indicates the repetition rate for the rear audio speakers.

PARK_INFO.ChimeType_Rear

This signal indicates the Chime Type for the rear audio speakers.

PARK_INFO.DisplayActivation

This signal indicates the request to display the car graphics layout on the IPC/EVIC screen.

STATUS_PAM.PAMSystemFault

This signal indicates if the PAM is in a faulted condition

STATUS_PAM.PAMSystemSts

This signal indicates the activation state of the PAM

STATUS_PAM.RearSensorSts

This signal indicates the activation state of the rear sensors

STATUS_PAM.PAM_LedControlSts

This signal indicates the request to illuminate the PAM LED status indicator

BH_IGW1.PamAlertMode

It indicates the selection of menu option to activate / deactivate visual alerts on the cluster display

LIN_BCM_IGW1.PAMRequestSts

It indicates the pressure of Park Assist pushbutton

STATUS_PAM.PAMAboveSpeed

This signal is used to manage a indication, related to the deactivation of PAM system, when the vehicle speed is too high.

TRANSM2.ShiftLeverPosition

This signal indicates the executed shift lever position.

BRAKE4.VehicleStandStillSts

This signal indicates if the car is in stand still condition

1.8.7 PROXI Parameters

The characteristics of the "PROXI" Parameters and their related values are defined in the "PROXI requirements" specific for the vehicle project.

CAN node 24 (PAM)

Gear_Box_Type

PAM_Configuration

1.9 Indication

PARKING ASSISTANCE.V1

Allocated on IPC

PARKING ASSISTANCE SENSOR BLOCKED.V1

Allocated on IPC

PARKING ASSISTANCE TEMPORARY FAIL.V1

Allocated on IPC

PARKING ASSISTANCE USER DISABLED.V1

Allocated on IPC

PARKING ASSISTANCE USER ENABLED.V1

Allocated on IPC

PARKING ASSISTANCE FAIL.V4

Allocated on IPC

PARKING ASSISTANCE DISABLED.V1

Allocated on IPC

PARKING ASSISTANCE UNAVAILABLE MAX SPEED.V1

Allocated on IPC

PARKING ASSISTANCE-VEHICLE-GRAPHIC.V2

Allocated on IPC

1.10 Working Conditions

1.10.1 Ignition Working Conditions Table

Ignition Working Conditions	Active Sub Function Names	Reference Paragraphs
Ignition_Off	-	-
Ignition ACC	-	-
Ignition_On	-	-
Ignition Pre Start	-	-
Ignition Start	-	-
Ignition Cranking	-	-
Ignition On Engine On	-	-
Ignition Pre ACC	-	-
Ignition Pre Off	-	-

1.10.2 Functional Working Conditions Table

Functional Working Conditions	Functional Requirements	Reference Paragraphs
Temporization After Ignition_Off	No	N.A.
Automatic Cranking (S&S)	No	N.A.
Automatic Stop (S&S)	No	N.A.
Low Battery	No	N.A.
Plant/Emission Mode	No	N.A.
Logistic Mode	No	N.A.

1.11 Functional Requirements

1.11.1 Parking Assistance System Management

1.11.1.1 PAM Requirements

1.11.1.1.1 Variable Definition

1.11.1.1.1.1 Reverse Condition

Reverse Condition is True **IF**:

the PROXI parameter Gear_Box_Type is different from "MTX" **AND** the signal *TRANSM2.ShiftLeverPosition* is equal to "R" for a time greater than TReverseGear.

1.11.1.1.1.2 Standstill Condition

Standstill Condition is True **IF**:

the signal *BRAKE4.VehicleStandStillSts* is equal to "True".

1.11.1.1.1.3 KEY ON Variable

The internal variable KEY ON is defined in order to indicate the following ignition states:

- Ignition_ON
- Ignition_Start
- Ignition_ON_Engine_ON

1.11.1.1.1.4 KEY OFF Variable

The internal variable KEY OFF is defined in order to indicate the following ignition states:

- Ignition_OFF
- Initialization

1.11.1.1.1.5 MODE_PAM Variable

The internal variable MODE_PAM is defined in order to store the last PAM activation state before the transition from KEY ON to KEY OFF. It can assume the following possible values:

- ON
- OFF

The initial value at Power ON is OFF

1.11.1.1.1.6 PAMFault Variable

The internal variable PAMFault is defined in order to store a failure status. It can assume the following possible values:

- TRUE: when a fault occurs
- FALSE: when no faults occur

1.11.1.1.2 PAM State Machine

The following State machine is defined in order to define the PAM Algorithm behaviour:

----- Create the State Machine -----

KEY OFF State Description

In the "KEY OFF" State,

- IF the signal *PAM_OperationalModeSts.Info* is equal to "Ignition_ON", "Ignition_Start", "Ignition_ON_Engine_ON"
THEN PAM shall move to "KEY ON" state.

KEY ON State Description

When PAM enters the "KEY ON" State,

- PAM shall turn on and perform initialization phase;
- PAM shall set all the signals present in STATUS_PAM and PARK_INFO to their default values, except for the signal *STATUS_PAM.PAMSystemSts* that shall be set to "ON_Inactive", as init value.

When PAM enters the "KEY ON" State,

- PAM shall wait a time equal to TFilter before considering the input signal *TRANSM2.ShiftLeverPosition* as valid
- during TFilter time, PAM shall consider the signal *TRANSM2.ShiftLeverPosition* equal to "P"

In the "KEY ON" State,

PAM shall load from its non volatile memory the last status, according to the internal variables MODE_PAM and PAMFault :

- IF the internal variable MODE_PAM is equal to "ON" AND the internal variable PAMFault is equal to "False"
THEN PAM shall:
move to the "ON_Enable" state
- IF the internal variable MODE_PAM is equal to "OFF" AND the internal variable PAMFault is equal to "False"
THEN PAM shall:
move to the "OFF" state
- IF the internal variable PAMFault is equal to "True"
THEN PAM shall:
move to the "Disable" state

In the "KEY ON" State,

- IF the signal *PAM_OperationalModeSts.Info* is equal to "Ignition_OFF" or "Initialization"
THEN PAM shall move to "KEY OFF" state.

PAM ON Enable State Description

When PAM enters the "ON_Enable" State, it shall

- set the internal variable PAMFault to "False"
- check if the Reverse Condition is verified:

- IF Reverse Condition is equal to *False*
THEN
PAM shall move to the "ON_Inactive" Substate.

- **IF** Reverse Condition is equal to *True*
THEN
PAM shall move to the "ON_Active" Substate.

In the "ON_Enable" state,

IF one of the following conditions is verified:

- (the signal *BRAKE1.VehicleSpeedVSOSig* becomes greater than *SPEED_LIMIT* **AND** Reverse Condition is equal to *True*) **OR**
- the hardwire signal *Trailer.Info* is equal to "*Present*" **OR**
- PAMFault becomes equal to "*True*" (see Diagnosis and Recovery Requirements)

THEN

PAM shall move to "Disable" state.

In the "ON_Enable" state,

IF the signal *LIN_BCM_IGW1.PAMRequestSts* switches to "*Pressed*"

THEN

PAM shall move to the "OFF" state.

- **PAM ON_Inactive SubState Description**

When PAM enters the "ON_Inactive" Substate, PAM shall:

- set the signal *STATUS_PAM.PAMSystemSts* to "*ON_Inactive*"
- set the signal *STATUS_PAM.RearSensorSts* to "*Active*"
- set the internal variable *MODE_PAM* to "ON" and store this value in its non volatile memory
- set the signal *STATUS_PAM.PAM_LedControlSts* equal to "*OFF*"
- set the signal *PARK_INFO.DisplayActivation* equal to "*Not Active*"

When PAM is in the ON_Inactive substate,

IF Reverse Condition becomes equal to *True*

THEN

PAM shall move to "ON_Active" substate.

- **PAM ON_Active SubState Description**

When PAM enters the "ON_Active" Substate, PAM shall:

- set the signal *STATUS_PAM.PAMSystemSts* to "*ON_Active*"
- set the signal *STATUS_PAM.RearSensorSts* to "*Active*"
- set the internal variable *MODE_PAM* to "ON" and store this value in its non volatile memory
- set the signal *STATUS_PAM.PAM_LedControlSts* equal to "*OFF*"
- set the signal *PARK_INFO.DisplayActivation* equal to "*Active*"

When PAM is in the "ON_Active" Substate, PAM shall:

set the signals of *PARK_INFO* message according to the table reported in *Visual Alerts Table* Paragraph.

- In the "ON_Active" substate, **IF**:
 - Standstill Condition is equal to "True" **AND**
 - an obstacle results continuously moving across two consecutive arcs

THEN

PAM shall:

- activate acoustic and visual warnings considering the closest position unless the obstacle is detected in a different zone continuously for *T_Filter_Alerts*

In the "ON_Active" substate,
IF Reverse Condition becomes equal to *False*
THEN
PAM shall move to "ON_Inactive" substate.

PAM Disable State Description

When PAM enters the "Disable" state, PAM shall:

- set the signal *STATUS_PAM.PAMSystemSts* to "ON_Disabled"
- set the signal *STATUS_PAM.PAM_LedControlSts* to "Continuous light"
- set the signal *STATUS_PAM.PAMAboveSpeed* to "True", only **IF** (the signal *BRAKE1.VehicleSpeedVSOSig* is greater than *SPEED_LIMIT* **AND** Reverse Condition is equal to *True*)

In the "Disable" state,

IF the signal *LIN_BCM_IGW1.PAMRequestSts* switches to "Pressed" **AND** the internal variable PAMFault is equal to "TRUE"

THEN

PAM shall:

- remain in the "Disable" state
- set the signal *STATUS_PAM.PAMSystemSts* equal to "Blinking light" for a time equal to *PAM_LED_BLINK_TIME* and then set the signal *STATUS_PAM.PAMSystemSts* equal to "Continuous light"

In the "Disable" state,

IF the signal *LIN_BCM_IGW1.PAMRequestSts* switches to "Pressed" **AND** the internal variable PAMFault is equal to "FALSE"

THEN

PAM shall move to the "OFF" state.

In the "Disable" state,

IF one of the following condition is verified:

- [(the signal *BRAKE1.VehicleSpeedVSOSig* becomes lower than *SPEED_LIMIT* **AND** Reverse Condition is equal to *True*) **AND** the internal variable PAMFault is equal to "False"] **OR**
- the hardwire signal *Trailer.Info* is different from "Present" **AND** the internal variable PAMFault is equal to "False" **OR**
- the internal variable PAMFault is equal to "False"

THEN

PAM shall move to "ON_Enable" substate.

PAM OFF State Description

When PAM enters the "OFF" state, PAM shall:

- set the signal *STATUS_PAM.PAMSystemSts* to "OFF"
- set the signal *STATUS_PAM.RearSensorSts* to "Active"
- set the internal variable *MODE_PAM* to "OFF" and store this value in its non volatile memory
- set the signal *STATUS_PAM.PAM_LedControlSts* to "Continuous light"

In the OFF state,

IF the signal *LIN_BCM_IGW1.PAMRequestSts* switches to "Pressed" **AND** the internal variable PAMFault is equal to "TRUE"

THEN

PAM shall:

- move to "Disable" state.

In the OFF state,

IF the signal *LIN_BCM_IOW1.PAMRequestSts* switches to "*Pressed*" **AND** the internal variable PAMFault is equal to "*FALSE*"

THEN

PAM shall move to "ON_Enable" state.

1.11.1.1.3 PAM - Chime Requirements

In the ON_Active state, PAM shall set the following signals:

- *PARK_INFO.ChimeActivation_LHR*
- *PARK_INFO.ChimeActivation_RHR*

according to the rules below:

Obstacle location	Signals value
Rear Left Zone	<i>PARK_INFO.ChimeActivation_LHR</i> equal to " <i>Active</i> " <i>PARK_INFO.ChimeActivation_RHR</i> equal to " <i>Not Active</i> "
Rear Central Zone	<i>PARK_INFO.ChimeActivation_LHR</i> equal to " <i>Active</i> " <i>PARK_INFO.ChimeActivation_RHR</i> equal to " <i>Active</i> "
Rear Right Zone	<i>PARK_INFO.ChimeActivation_LHR</i> equal to " <i>Not Active</i> " <i>PARK_INFO.ChimeActivation_RHR</i> equal to " <i>Active</i> "

PAM shall:

- set the signal *PARK_INFO.ChimeType_Rear* to "*Type4*"
- set the signal *PARK_INFO.ChimeRepetitionRate_Rear* according to the following rules:
 - *repetition rate* shall vary linearly with the obstacle distance
 - *pulse duration* between two chimes shall assume variable values between REP_MIN_DURATION and REP_MAX_DURATION
 - *continuous tone* shall correspond to the value 0

WALL DETECTION

IF following devices:

- *R_Bumper_Left_Parking.Sensor*
- *R_Bumper_Right_Parking.Sensor*

detect "wall shaped obstacle" (see definition in FTS document) continuously for more than Tpmwalldet

THEN

PAM shall deactivate the acoustic feedback as long as the obstacle is detected.

MULTIPLE OBSTACLES - CHIME MANAGEMENT

1) **IF** more obstacles are detected in the same zone,

THEN

only the nearest obstacle shall be acoustically signalled.

2) **IF** more obstacles are detected in different zones, in the Rear Detection Area

THEN

the nearest obstacle shall be acoustically signalled

1.11.1.2 LSS Requirements

1.11.1.2.1 Switch Acquisition

IF the *Park_Assist_On_Off.Switch* is pressed and the signal *Park_Assist_On_Off_Switch.Reg* switches from "Not Pressed" to "Pressed"

THEN

LSS shall set the hardwire signal *Park_Assist_On_Off.Reg* equal to "Pressed"

IF the *Park_Assist_On_Off.Switch* is released and the signal *Park_Assist_On_Off_Switch.Reg* switches from "Pressed" to "Not Pressed"

THEN

LSS shall set the hardwire signal *Park_Assist_On_Off.Reg* equal to "Not Pressed"

1.11.1.2.2 LED Management

IF the hardwire signal *LedControlSts.Reg* is equal to "Off"

THEN

LSS shall set the internal signal *Park_Assist_Light.Cmd* to "Off"

IF the hardwire signal *LedControlSts.Reg* is equal to "Continuous Light_DAY"

THEN

LSS shall set the internal signal *Park_Assist_Light.Cmd* to "Continuous Light_DAY"

IF the hardwire signal *LedControlSts.Reg* is equal to "Continuous Light_NIGHT"

THEN

LSS shall set the internal signal *Park_Assist_Light.Cmd* to "Continuous Light_NIGHT"

IF the hardwire signal *LedControlSts.Reg* is equal to "Blinking Light_DAY"

THEN

LSS shall set the internal signal *Park_Assist_Light.Cmd* to "Blinking Light_DAY"

IF the hardwire signal *LedControlSts.Reg* is equal to "Blinking Light_NIGHT"

THEN

LSS shall set the internal signal *Park_Assist_Light.Cmd* to "Blinking Light_NIGHT"

1.11.1.3 BCM Algorithm Requirements

1.11.1.3.1 PAM button and LED Management

The following requirements shall be implemented only if the PROXI parameter Can Node 24 is set to "Present"

IF the signal *BCM_OperationalModeSts.Info* is equal to "Ignition_On" or "Ignition_Start" or "Ignition_On_Engine_On"

THEN

- **IF** the hardwire signal *Park_Assist_On_Off.Reg* is equal to "Pressed"

THEN

BCM shall set the signal *LIN_BCM_IGW1.PAMRequestSts* equal to "Pressed"

- **IF** the hardwire signal *Park_Assist_On_Off.Reg* is equal to "Not Pressed"

THEN

BCM shall set the signal *LIN_BCM_IGW1.PAMRequestSts* equal to "Not Pressed"

IF the signal *BCM_OperationalModeSts.Info* is equal to "Ignition_On" or "Ignition_Start" or "Ignition_On_Engine_On"

THEN

- **IF** the signal *STATUS_PAM.PAM_LedControlSts* is equal to "Off"

THEN

BCM shall set the hardwire signal *LedControlSts.Reg* equal to "Off"

- **IF** the signal *STATUS_PAM.PAM_LedControlSts* is equal to "Continuous Light" **AND** the internal signal *InternalLightSts* is equal to "Not Active",

THEN

BCM shall set the hardwire signal *LedControlSts.Req* equal to "Continuous Light_DAY" according to the "Technical Esthetic Finalized Specification" document.

- **IF** the signal *STATUS_PAM.PAM_LedControlSts* is equal to "Continuous Light" **AND** the internal signal *InternalLightSts* is equal to "Active",

THEN

BCM shall set the hardwire signal *LedControlSts.Req* equal to "Continuous Light_NIGHT" according to the "Technical Esthetic Finalized Specification" document.

- **IF** the signal *STATUS_PAM.PAM_LedControlSts* is equal to "Blinking Light" **AND** the internal signal *InternalLightSts* is equal to "Not Active",

THEN

BCM shall set the hardwire signal *LedControlSts.Req* equal to "Blinking Light_DAY" with a duty cycle equal to *PAM_LED_BLINK_DUTY* and a frequency equal to *PAM_LED_BLINK_FREQ* according to the "Technical Esthetic Finalized Specification" document.

- **IF** the signal *STATUS_PAM.PAM_LedControlSts* is equal to "Blinking Light" **AND** the internal signal *InternalLightSts* is equal to "Active",

THEN

BCM shall set the hardwire signal *LedControlSts.Req* equal to "Blinking Light_NIGHT" with a duty cycle equal to *PAM_LED_BLINK_DUTY* and a frequency equal to *PAM_LED_BLINK_FREQ* according to the "Technical Esthetic Finalized Specification" document.

1.12 Diagnosis and Recovery

1.12.1 Diagnosis Table

ID	Component / Signal / Message Name	Diagnosis Type	Functional Recovery	DTC Stored	Detect Condition	Fault Signal
1.0	Sensors	SC, OC	PAM	PAM	"Ignition_ON" or "Ignition_On_Engine_On"	<i>STATUS_PAM.PAMSystemFault</i>
2.0	PAM	Internal fail	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
3.0	<i>BRAKE1</i>	Missing Message	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
4.0	<i>STATUS_PAM</i>	Missing Message	IPC	IPC	Ignition_ON Ignition_On_Engine_On	N.A.
5.0	<i>PAM_OperationalModeSts.Info</i>	Plausibility	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
6.0	Sensors	Plausibility	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.RearSensorSts</i>

7.0	<i>STATUS_TELEMATIC</i>	Missing Message	IPC	IPC	Ignition_ON Ignition_On_Engine_On	N.A.
8.0	<i>LIN_BCM_IGW1</i>	Missing Message	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
9.0	<i>BH_IGW1</i>	Missing Message	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
10.0	<i>LIN_BCM_IGW1.PAMRequestSts</i>	Plausibility	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
11.0	<i>BRAKE1.VehicleSpeedVSO Sig BRAKE1.VehicleSpeedVSO SigFailSts</i>	Plausibility	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
12.0	PAM	Node Mute	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
13.0	PAM	Node Bus Off	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
14.0	<i>STATUS_PAM.PAMLedControlSts</i>	Plausibility	BCM	BCM	Ignition_ON Ignition_On_Engine_On	N.A.
15.0	<i>STATUS_PAM</i>	Missing Message	BCM	BCM	Ignition_ON Ignition_On_Engine_On	N.A.
16.0	PAM	Configuration Missing or Calibration Mismatch	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
17.0	<i>TRANSM2</i>	Missing Message	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
18.0	<i>TRANSM2.ShiftLeverPosition</i>	Plausibility	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
19.0	<i>BRAKE4</i>	Missing Message	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>
20.0	<i>BRAKE4.VehicleStandStillSts</i>	Plausibility	PAM	PAM	Ignition_ON Ignition_On_Engine_On	<i>STATUS_PAM.PAMSystemFault</i>

1.12.2 Diagnosis Table and Recovery Description

In case of systems equipped with diagnosis capabilities, the supplier developed functions shall be compliant with the company standards.

Following requirements have to be considered as sub-set diagnosis requirements (usually correlated to the system functionality).

Any other specific diagnosis requirement is included in the "PAM - Diagnostic Requirements" document for the system itself.

1.12.2.1 PAM Requirements

1.12.2.1.1 ID 1.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM detects a failure in one of the following devices:

- **R_Bumper_Left_Parking.Sensor**
- **R_Bumper_Right_Parking.Sensor**
- **R_Bumper_Central_Parking.Sensor**

for the period indicated in "PAM - Diagnostic Requirements" document,

THEN

PAM shall:

- set the proper DTC as long as the failure persists, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "*True*" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** the PAM is not in the "*OFF*" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set the signal *STATUS_PAM.PAMSystemFault* to "*System_Failure*"
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "*False*"
- move to the "ON_Enable" state

1.12.2.1.2 ID 2.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM detects an failure for the period indicated in "PAM - Diagnostic Requirements" document

THEN

PAM shall:

- set the proper DTC as long as the failure persists, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "*True*" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** the PAM is not in the "*OFF*" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set the signal *STATUS_PAM.PAMSystemFault* to "*System_Failure*"
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "*False*"
- move to the "ON_Enable" state

1.12.2.1.3 ID 3.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM detects Missing Message of "*BRAKE1*" for the period indicated in "PAM - Diagnostic Requirements"

THEN

PAM shall:

- set the proper DTC as long as the failure persists, according to "PAM - Diagnostic Requirements" document
- set the internal variable *PAMFault* equal to "*True*" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** the PAM is not in the "*OFF*" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set the signal *STATUS_PAM.PAMSystemFault* to "*External_Failure*"
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "*False*"
- move to the "ON_Enable" state

1.12.2.1.4 ID 5.0 Description

IF PAM receives the signal *PAM_OperationalModeSts.Info* equal to "*SNA*" for the period indicated in "PAM - Diagnostic Requirements" document

THEN

PAM shall:

- set the proper DTC, according to "PAM - Diagnostic Requirements" document
- set the internal variable *PAMFault* equal to "*True*" and store this value in its non volatile memory
- disable the Parking Assistance functionality
- set the signal *STATUS_PAM.PAMSystemFault* to "*System_Failure*"
- move to the "Disabled" state

IF PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "False"
- move to the "ON_Enable" state

1.12.2.1.5 ID 6.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" AND PAM detects a blockage in one of the following devices

- **R_Bumper_Left_Parking.Sensor**
- **R_Bumper_Right_Parking.Sensor**
- **R_Bumper_Central_Parking.Sensor**

for the period indicated in "PAM - Diagnostic Requirements" document and with a vehicle speed between Min_Speed and Max_Speed:

THEN

PAM shall:

- set the proper DTC as long as the failure persists, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" AND the PAM is not in the "OFF" state

THEN

PAM shall:

- set the signal *STATUS_PAM.RearSensorSts* to "Not_Active_Blocked"
- set the signal *STATUS_PAM.PAMSystemFault* to "External_Failure"
- disable the Parking Assistance functionality
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" AND PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.RearSensorSts* to "Active"
- move to the "ON_Enable" state

1.12.2.1.6 ID 8.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" AND PAM detects Missing Message of the message "LIN_BCM_IGWI" for the period indicated in "PAM - Diagnostic Requirements"

THEN

PAM shall:

- set the proper DTC as long as the missing message occurs, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "*External_Failure*"
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "*False*"
- move to the "ON_Enable" state

1.12.2.1.7 ID 9.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM detects Missing Message of the message "*BH_IGW1*" for the period indicated in "PAM - Diagnostic Requirements"

THEN

PAM shall:

- set the proper DTC as long as the missing message occurs, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "*True*" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "*External_Failure*"
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "*False*"
- move to the "ON_Enable" state

1.12.2.1.8 ID 10.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "*Ignition_ON*" or "*Ignition_On_Engine_On*" **AND** PAM detects that the signal *LIN_BCM_IGW1.PAMRequestSts* doesn't switch from "*Pressed*" to "*Not Pressed*" within *PAM_STUCK_TIMEOUT*,

THEN

PAM shall:

- set the proper DTC as long as the failure occurs, according to "PAM - Diagnostic Requirements" document

• set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "External_Failure"
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "False"
- move to the "ON_Enable" state

1.12.2.1.9 ID 11.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" **AND** PAM receives the signals:

- *BRAKE1.VehicleSpeedVSOSig* is equal to "SNA" **OR**
- *BRAKE1.VehicleSpeedVSOSigFailSts* is equal to "Fail Present"

THEN

PAM shall:

- set the proper DTC as long as the failure occurs, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "External_Failure"
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "False"
- move to the "ON_Enable" state

1.12.2.1.10 ID 12.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" and PAM is in condition of Node Mute

THEN

PAM shall:

- set the proper DTC as long as the failure occurs, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "True" and store this value in its non volatile memory
- disable the Parking Assistance functionality

1.12.2.1.11 ID 13.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" and PAM is in condition of Node Bus Off

THEN

PAM shall:

- set the proper DTC as long as the failure occurs, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "True" and store this value in its non volatile memory
- disable the Parking Assistance functionality

1.12.2.1.12 ID 16.0 Description

IF PAM detects an error of "Vehicle Configuration Missing" or an error of "Vehicle Calibration Mismatch" on *PAM_Tuning_Set PROXI* parameter

THEN

PAM shall:

- Set an appropriate DTC according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "System_Failure"
- move to the "Disabled" state

IF PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "False"
- move to the "ON_Enable" state.

1.12.2.1.13 ID 17.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" and PAM detects Missing Message of "TRANSM2" for the period indicated in "PAM - Diagnostic Requirements"

THEN

PAM shall:

- Set an appropriate DTC according to "PAM - Diagnostic Requirements" document

• set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "External_Failure"
- move to the "Disabled" state

IF PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "False"
- move to the "ON_Enable" state.

1.12.2.1.14 ID 18.0 Description

This diagnosis shall be implemented IF PROXI parameter Gear_Box_Type is different from 'MTX'.

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" and PAM receives

- the signal *TRANSM2.ShiftLeverPosition* equal to "SNA"

THEN

PAM shall:

- Set an appropriate DTC according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "External_Failure"
- move to the "Disabled" state

IF PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "False"
- move to the "ON_Enable" state.

1.12.2.1.15 ID 19.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" OR "Ignition_ON_Engine_ON" and PAM detects Missing Message of "BRAKE4" for the period indicated in "PAM - Diagnostic Requirements"

THEN

PAM shall:

- set the proper DTC as long as the Missing Message occurs, according to "PAM - Diagnostic Requirements" document

• set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" OR "Ignition_ON_Engine_ON" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "External_Failure"
- move to the "Disabled" state

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" OR "Ignition_ON_Engine_ON" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "False"
- move to the "ON_Enable" state

1.12.2.1.16 ID 20.0 Description

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" OR "Ignition_ON_Engine_ON" and PAM receives the signal *BRAKE4.VehicleStandStillSts* equal to "SNA"

THEN

PAM shall:

- set the proper DTC as long as the failure occurs, according to "PAM - Diagnostic Requirements" document
- set the internal variable PAMFault equal to "True" and store this value in its non volatile memory

When the failure occurs, **IF** *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" OR "Ignition_ON_Engine_ON" **AND** the PAM is not in the "OFF" state

THEN

PAM shall:

- disable the Parking Assistance functionality
- set *STATUS_PAM.PAMSystemFault* to "External_Failure"
- move to the "Disable" state

IF *PAM_OperationalModeSts.Info* is equal to "Ignition_ON" OR "Ignition_ON_Engine_ON" **AND** PAM doesn't detect this failure anymore

THEN

PAM shall:

- Heal the DTC
- set the signal *STATUS_PAM.PAMSystemFault* to "False"
- move to the "ON_Enable" state

1.12.2.2 IPC Requirements

ID 4.0 Description

IPC shall perform the following diagnosis only if the PROXI parameter Can Node 24 is set to "Present".

IF *IPC_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" and the IPC detects Missing Message of *STATUS_PAM*

THEN

IPC shall:

- activate **PARKING ASSISTANCE FAIL indication**
- set internal signal DisplayActivation to "NotActive"
- set a specific DTC according to "IPC - Diagnostic Requirements" document as long as the fault persists.

IF *IPC_OperationalModeSts.Info* is equal to "Ignition_ON" OR "Ignition_ON_Engine_ON" and the IPC receives again the message *STATUS_PAM*

THEN

IPC shall:

- heal the DTC according to "IPC - Diagnostic Requirements" document
- resume normal functioning

ID 7.0 Description

IPC shall perform the following diagnosis only if the PROXI parameter Can Node 24 is set to "Present".

IF IPC does not receive the message *STATUS_TELEMATIC* continuously for T_telematic

THEN

IPC shall:

- set a specific DTC according to "IPC - Diagnostic Requirements" document
- manage the **PARKING ASSISTANCE.V1** indication, according to "Indication Management" paragraph.

IF IPC receives again the message *STATUS_TELEMATIC* continuously for T_Telematic

THEN

IPC shall:

- heal the DTC according to "IPC - Diagnostic Requirements" document
- stop managing the **PARKING ASSISTANCE.V2** indication.

1.12.2.3 BCM Requirements

BCM shall perform the following diagnosis only if the PROXI parameter Can Node 24 is set to "Present".

ID 14.0 Description

IF *BCM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" and BCM receives

- the signal *STATUS_PAM.PAM_LedControlSts* equal to "SNA" for the period DTC_BCM

THEN

BCM shall:

- set a specific DTC as long as failure persists according to "BCM - Diagnostic Requirements" document
- set Park_Assist_Light.Cmd to the previous value until BCM receives a valid value different from "SNA"

IF BCM doesn't detect this failure anymore

THEN

BCM shall:

- heal the DTC according to "BCM- Diagnostic Requirements" document

ID 15.0 Description

IF *BCM_OperationalModeSts.Info* is equal to "Ignition_ON" or "Ignition_On_Engine_On" and BCM detects Missing Message of *STATUS_PAM* for the period DTC_BCM

THEN

BCM shall:

- set a specific DTC as long as Missing Message occurs, according to "BCM - Diagnostic Requirements" document
- set Park_Assist_Light.Cmd to the previous value as long as Missing Message occurs.

IF BCM does not detect this failure anymore

THEN

BCM shall:

- heal the DTC according to "BCM- Diagnostic Requirements" document

1.12.3 Diagnostic Requirements

1.12.3.1 PAM requirements

1.12.3.1.1 DTC enabling conditions

Before detecting all the faults related to CAN messages, PAM shall wait a time equal to T_{ign} (**this parameter shall be tunable**) exiting from the initialisation phase.

1.12.3.1.2 Engineering parameters

A specific parameter in the diagnostic section of the ECU shall be implemented to indicate the presence of a trailer hook if one is installed.

In case of failure detection, PAM shall store an appropriate DTC in its internal memory and freeze, among environmental parameters, vehicle speed detected at the fault detection (information available in the signal *VehicleSpeedVSOSig*. If *VehicleSpeedVSOSigFailSts* is set to "Fail Present", then PAM shall store default value of *VehicleSpeedVSOSig*.

1.13 Configuration Parameters

1.13.1 Configuration Parameters Table

ID	Parameter Name	First Trial Value	Range Value	Resolution	Unit	Component	Implementation	XCP Availability	Impact	Parameter Description
1	T_{ign}	2000	$[0 \div 3000]$	100	ms	PAM	SUPPLIER_EOL_2	No	Diagnosis	Filter time before enabling diagnosis
2	REP_MIN_DURATION	0	$[0 \div 400]$	25	ms	PAM	DEFINE	No	N.A.	Minimum allowed duration for chime repetition rate

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3	REP_MAX_DURATION	375	[0 ÷ 400]	25	ms	PAM	DEFINE	No	N.A.	Maximum allowed duration for chime repetition rate
4	ARC_FLASH_DUTY_CYCLE	50	[0;100]	5	%	IPC	DEFINE	No	N.A.	Duty cycle of flashing. See IPC specification for definition
5	ARC_FLASH_PERIOD	600	[0;1000]	50	ms	IPC	SUPPLIER_EOL_2	No	N.A.	Period of flashing cycle of arcs. See IPC specification for definition
6	PAM_LED_BLINK_DUTY	50	[0 ÷ 100]	50	%	BCM	DEFINE	No	N.A.	Duty cycle of LED blinking
7	PAM_LED_BLINK_FREQ	2	[0 ÷ 10]	2	Hz	BCM	SUPPLIER_EOL_2	No	N.A.	Frequency of LED blinking
8	PAM_STUCK_TIME_OUT	15	[0 ÷ 30]	1	s	PAM	SUPPLIER_EOL_2	No	N.A.	
9	PAM_LED_BLINK_TIME	5	[0 ÷ 10]	1	s	PAM	SUPPLIER_EOL_2	No	N.A.	Time of LED blinking
10	TReverseGear	400	[0 ÷ 1000]	10	ms	PAM	SUPPLIER_EOL_2	No	N.A.	
11	SPEED_LIMIT	11	[0 ÷ 100]	1	km/h	PAM	SUPPLIER_EOL_2	No	N.A.	
12	Min_Speed	2	[0 ÷ 100]	1	km/h	PAM	SUPPLIER_EOL_2	No	N.A.	
13	Max_Speed	50	[0 ÷ 100]	1	km/h	PAM	SUPPLIER_EOL_2	No	N.A.	
14	Tpamwalldet	3	[0 ÷ 10]	1	s	PAM	SUPPLIER_EOL_2	No	N.A.	

15	TFilter	1000	[0 ÷ 5000]	100	ms	PAM	SUPPLIER_EOL_2	No	N.A.	Filter time after the initialization phase
16	T_PAMSts_valid	1000	[0 ÷ 3000]	500	ms	IPC	SUPPLIER_EOL_2	No	N.A.	
17	T_Filter_Alerts	1000	[0 ÷ 2000]	100	ms	PAM	SUPPLIER_EOL_2	No	N.A.	Filter time to update the warnings
18	T_Telematic	3000	[0 ÷ 5000]	1000	ms	IPC	SUPPLIER_EOL_2	No	N.A.	Filter time to validate the condition of unavailability of the radio

The symbol "***" means: "See column Parameter Description".

1.13.2 Configuration Parameters Table Description

1.13.2.1 ID 906 Description

When PAM receives Vehicle_Line_Configuration with a valid enumeration, then PAM shall select proper parameter set for the vehicle model.

1.14 Wake Up Stay Active

1.14.1 Wake Up and Stay Active Event Table

ID	Signal Name	Event Type	Component Awoken / Waking up	Wake up Type	Waking Up Functionality
N.A.	N.A.	N.A.	N.A.	N.A.	N.A.

1.14.2 Wake Up and Stay Active Event Table Description

Not Applicable

1.15 Acronyms and Glossary

SC:	Short Circuit
OC:	Open Circuit
EOL:	End Of Line

1.16 Reference Documents

PAM - System Technical Specification

IPC - System Technical Specification

PAM - Diagnostic Requirements

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